Juan RESTREPO

Full-Stack AI Engineer

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*Full-stack engineer with 3 years of experience building **applications that help businesses monitor and manage their operations in real time** (remote monitoring, alerts, dashboards), adapting this expertise to each industry's needs. I use **React, Python and Azure**, and I build software by orchestrating **AI agents (Claude Code, MCP)**, from spec to production. Fluent in **French, English, Spanish and Portuguese**.*

Professional Experience

Full-Stack AI Engineer *(January 2026 – present)*

Olenergies, Bagnolet, France · Energy start-up (smart batteries replacing diesel generators) · B2B clients

- **Web monitoring platform:** Built independently with **AI agents**; clients and their operators now track all their installations live, where they previously had no visibility. It has become a key selling point against competitors.
- **Remote console:** Secure access from the web platform to each installation's touchscreen (multi-site architecture, no equipment exposed to the internet): site visits now happen only when needed.
- **Team AI adoption:** Brought AI agents (**Claude Code**) into the team's daily workflow, with shared project rules and automated code review: faster delivery and consistent quality.
- **Quality & CI/CD delivery: GitHub Actions** pipelines and deployment to **Azure**: weekly, fully tested releases with zero downtime.

Full-Stack & IoT Engineer (work-study) *(2023 – 2025)*

Olenergies, Montreuil, France

- **Embedded touchscreen: React** interface installed on each battery, showing solar production, charge level and alarms in real time: each installation can be operated and diagnosed on site.
- **Backend & REST API: Python (FastAPI)** backend collecting equipment data (**MQTT**) and streaming it live to the dashboards (**REST API**, **WebSocket**), refreshed every 1–2 seconds.
- **Product lifecycle:** End-to-end involvement across the product lifecycle (specifications, testing on test benches and live installations, deployment and maintenance).

Professional & Personal Projects — live demos at juancrestrepo.dev

OlenPeps · *professional project (Olenergies) · live demo* — Lets businesses track their solar and battery installations wherever they are: battery level, solar production and alarms, live on a map.

MoviesApp · *personal project · live demo* — Helps you choose what to watch: a movie and TV catalog with streaming platforms by country, and an AI assistant (Claude API, tool calling) that recommends titles from real catalog data.

Technical Skills

AI & agents: Claude Code, Claude API (LLM), tool calling, MCP, spec-driven development.

Frontend: React, TypeScript, TanStack Query, Tailwind.

Backend: Python, FastAPI, PostgreSQL, SQLAlchemy, Azure SQL (SQL Server), InfluxDB, WebSocket, MQTT.

Infra & cloud: Docker, Azure, Vercel, Linux, Nginx, WireGuard, Grafana, Sentry.

Quality, CI/CD & methods: Git, GitHub Actions, Pytest, Vitest, Testing Library, Agile (Scrum).

Education

Master's Degree in Engineering (work-study) *(2023 – 2025)*

ENSEA, Cergy, France · Computer Science track

Bachelor's Degree in Electronic Engineering *(2019 – 2022)*

Universidad Tecnológica de Bolívar, Colombia · "Talento Caribe" excellence scholarship (100%) · Erasmus+ exchange at ENSEA (2023)

Languages

Spanish (native) • French (C1) • English (C1) • Portuguese (C1) • Italian (B1)